

4.2 Double Integrals over Nonrectangulr Regions_Guide

Monday, January 12, 2026 10:42 PM

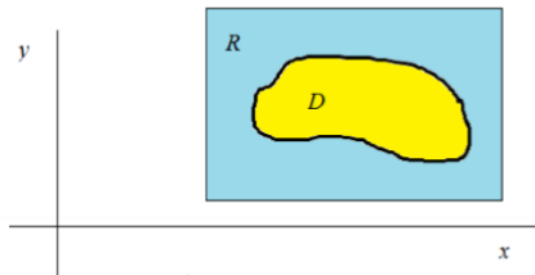


4.2 Double
Integrals ...

4.2 Double Integrals over Nonrectangular Regions

Motivation: We want to be able to integrate two variable functions not just over rectangles but also over regions D of a more general shape.

Suppose we want to integrate the function $f(x, y)$ over a bounded region D . Since D is bounded we know that it can be enclosed in a rectangle. A picture of what D might look like along with a rectangle it is contained in is:



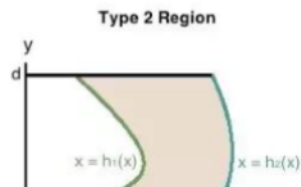
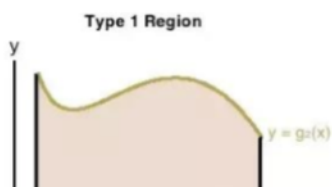
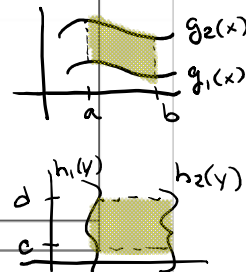
We define $F(x, y) = \begin{cases} f(x, y) & \text{if } (x, y) \text{ is in } D \\ 0 & \text{if } (x, y) \text{ is NOT in } D \text{ but is in } R \end{cases}$

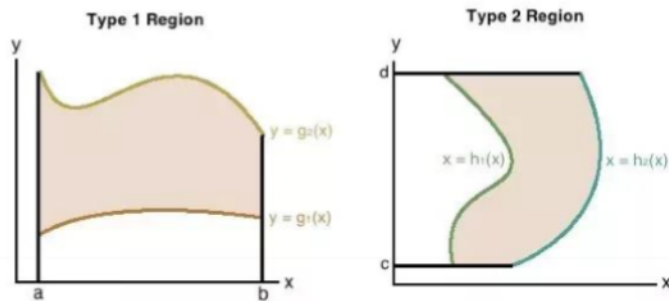
So we expect $\iint_R F(x, y) dA = \iint_D f(x, y) dA$

Definition: If F is integrable over R , then we define the double integral of f over D by $\iint_D f(x, y) dA = \iint_R F(x, y) dA$ (where we are using the notation and symbols introduced in the motivation).

Definition: We say that a plane region D is a

- type 1 region if it lies between the graphs of two continuous functions of x , that is, $D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ where $g_1(x)$ and $g_2(x)$ are continuous on $[a, b]$.
- type 2 region if it lies between the graphs of two continuous functions of y , that is, $D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ where $h_1(y)$ and $h_2(y)$ are continuous on $[c, d]$.





Exploration:

Suppose we wish to evaluate $\iint_D f(x,y) dA$ when $D = \{(x,y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ is a type 1 region and f is continuous on D . To do this, we pick a rectangle $R = [a,b] \times [c,d]$ containing D . A picture of what R might look like is shown below.

$$\text{Let } F(x,y) = \begin{cases} f(x,y) & \text{if } (x,y) \text{ is in } D \\ 0 & \text{if } (x,y) \text{ is in } R \text{ but not in } D \end{cases}$$

$$\text{Then } \iint_D f(x,y) dA = \iint_R F(x,y) dA$$

By Fubini's Thm we know

$$\iint_R F(x,y) dA = \int_a^b \left[\int_c^d F(x,y) dy \right] dx$$

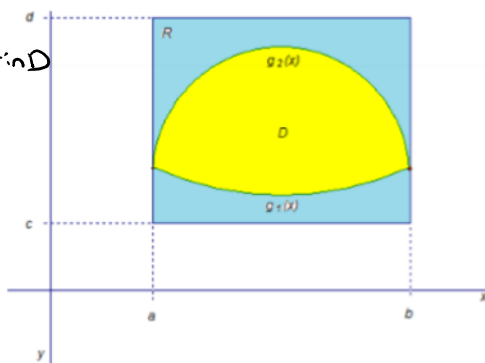
Since $F(x,y) = 0$ when $y > g_2(x)$ or $y < g_1(x)$ & $F(x,y) = f(x,y)$ when (x,y) is in D

$$\int_c^d F(x,y) dy = \int_c^{g_1(x)} 0 dy + \int_{g_1(x)}^{g_2(x)} f(x,y) dy + \int_{g_2(x)}^d 0 dy$$

$$= 0 + \int_{g_1(x)}^{g_2(x)} f(x,y) dy + 0$$

$$= \int_{g_1(x)}^{g_2(x)} f(x,y) dy$$

$$\text{So } \iint_D f(x,y) dA = \iint_R F(x,y) dA = \int_a^b \left[\int_{g_1(x)}^{g_2(x)} f(x,y) dy \right] dx$$



We now summarize the results of the exploration above.

Theorem:

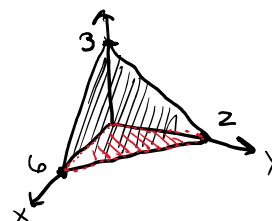
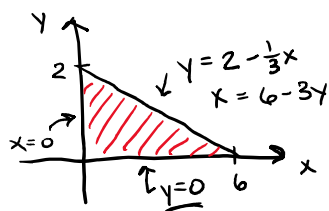
1. If f is continuous on a *type 1* region D such that $D = \{(x, y) \mid a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$ and both $g_1(x)$ and $g_2(x)$ are continuous on $[a, b]$, then $\iint_D f(x, y) dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) dy dx$.
2. If f is continuous on a *type 2* region D such that $D = \{(x, y) \mid c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\}$ and both $h_1(y)$ and $h_2(y)$ are continuous on $[c, d]$, then $\iint_D f(x, y) dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) dx dy$.

Question: Could we interchange the order of the integrals and differentials in the most recent theorem?

Example 1: Find the volume of the solid which is in the first octant and lies under the plane given by:
 $x + 3y + 2z = 6$.

First note we want the volume under
 $x + 3y + 2z = 6 \rightarrow z = 3 - \frac{1}{2}x - \frac{3}{2}y$

& we are integrating over a region D



Type 1: $D = \{(x, y) \mid 0 \leq x \leq 6 \text{ \& } 0 \leq y \leq 2 - \frac{1}{3}x\}$
 OR

Type 2: $D = \{(x, y) \mid 0 \leq y \leq 2 \text{ \& } 0 \leq x \leq 6 - 3y\}$

$$\text{So } V = \iint_D (3 - \frac{1}{2}x - \frac{3}{2}y) dA = \int_0^2 \int_0^{6-3y} (3 - \frac{1}{2}x - \frac{3}{2}y) dx dy$$

$$= \int_0^2 \left[3x - \frac{1}{4}x^2 - \frac{3}{2}xy \right]_0^{6-3y} dy$$

$$= \int_0^2 \left[3(6-3y) - \frac{1}{4}(6-3y)^2 - \frac{3}{2}(6-3y)y \right] dy$$

⋮

$$= \int_0^2 \left[9 - 9y + \frac{9}{4}y^2 \right] dy$$

$$= 9y - \frac{9}{2}y^2 + \frac{3}{4}y^3 \Big|_0^2$$

$$= 9y - \frac{9}{2}y^2 + \frac{3}{4}y^3 \Big|_0^2$$

$$= (18 - 18 + 6) - (0)$$

$$= 6$$

Question: Why would it be incorrect to use the iterated integral $\int_0^2 \left[\int_0^6 \frac{6-3y-x}{2} dx \right] dy$ to find the desired volume in this example?

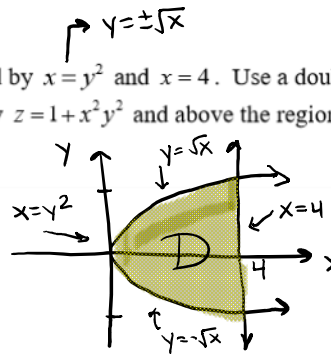
Example 2: Suppose that D is the region in the xy -plane bounded by $x = y^2$ and $x = 4$. Use a double integral to find the volume of the solid that lies below the surface given by $z = 1 + x^2 y^2$ and above the region D .

We first sketch a picture of D .

$$D = \{(x, y) \mid 0 \leq x \leq 4 \text{ \& \; } -\sqrt{x} \leq y \leq \sqrt{x}\}$$

OR

$$D = \{(x, y) \mid -2 \leq y \leq 2 \text{ \& \; } y^2 \leq x \leq 4\}$$



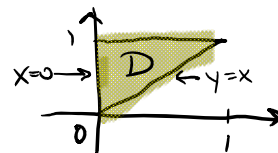
$$V = \iint_D (1 + x^2 y^2) dA = \int_{-2}^2 \int_{y^2}^4 (1 + x^2 y^2) dx dy$$

Example 3: Evaluate the iterated integral $\int_0^1 \int_x^1 \sin(y^2) dy dx$.

$\sin(y^2)$ has no easy antiderivative,

Instead let's try to figure out the region D , & then redefine D so we can integrate w/ respect to x first.

Note $D = \{(x, y) \mid \underbrace{x \leq y}_{y=x} \leq 1 \text{ \& } 0 \leq x \leq 1\}$



let $D = \{(x, y) \mid 0 \leq y \leq 1 \text{ \& } 0 \leq x \leq y\}$

$$\begin{aligned} \text{So } \int_0^1 \int_x^1 \sin(y^2) dy dx &= \int_0^1 \int_0^y \sin(y^2) dx dy = \int_0^1 x \sin(y^2) \Big|_0^y dy \\ &= \int_0^1 y \sin(y^2) dy = \int_0^1 \frac{1}{2} \sin(u) du = -\frac{1}{2} \cos(u) \Big|_0^1 = -\frac{1}{2} \cos(1) + \frac{1}{2} \cos(0) \\ &= -\frac{1}{2} \cos(1) + \frac{1}{2} \end{aligned}$$

$u = y^2$
 $du = 2y dy \rightarrow \frac{1}{2} du = y dy$

Example 4: Evaluate $\iint_D x e^{y^3} dA$ over the region D in the first quadrant which is bounded by the graphs $y = x^2$, $x = 0$, and $y = 4$.

Double Integral Properties Over a Region D :

Assume that $\iint_D f(x, y) dA$ and $\iint_D g(x, y) dA$ exist. Also suppose that c is a constant. Then,

1. $\iint_D [f(x, y) + g(x, y)] dA = \iint_D f(x, y) dA + \iint_D g(x, y) dA$
2. $\iint_D cf(x, y) dA = c \iint_D f(x, y) dA$
3. If $f(x, y) \geq g(x, y)$ for all (x, y) in D , then $\iint_D f(x, y) dA \geq \iint_D g(x, y) dA$
4. If $D = D_1 \cup D_2$ where D_1 and D_2 don't overlap except perhaps at the boundaries, then $\iint_D f(x, y) dA = \iint_{D_1} f(x, y) dA + \iint_{D_2} f(x, y) dA$.
5. If $A(D)$ represents the area of D , then $\iint_D 1 dA = A(D)$.
6. If $m \leq f(x, y) \leq M$ for all $(x, y) \in D$, then $mA(D) \leq \iint_D f(x, y) dA \leq MA(D)$.

Example 5: Suppose that D is the region in the xy -plane bounded by $x = y^2$ and $x = 4$. Use a double integral to find the area of D .